

- Antihistaminic actions may cause sedation, weight gain
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Mechanism of weight gain and any possible increased incidence of diabetes or dyslipidemia with conventional antipsychotics is unknown

Notable Side Effects

- * Neuroleptic-induced deficit syndrome
- * Akathisia
- * Drug-induced parkinsonism
- * Tardive dyskinesia
- * Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- * Galactorrhea, amenorrhea
- Dizziness, sedation
- Dry mouth, constipation, urinary retention, blurred vision
- Decreased sweating
- Sexual dysfunction
- Hypotension, tachycardia, syncope
- Weight gain



Life-Threatening or Dangerous Side Effects

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare jaundice, agranulocytosis
- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain



- Many experience and/or can be significant in amount

Sedation



- Many experience and/or can be significant in amount
- Sedation is usually transient

What to Do About Side Effects

- Wait
- Wait
- Wait

- For drug-induced parkinsonism, add an anticholinergic agent
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Reduce the dose
- For sedation, give at night
- Switch to an atypical antipsychotic
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- Psychosis: oral: 12–24 mg/day; 16–64 mg/day in hospitalized patients
- Nausea/vomiting: 8–16 mg/day oral, 5 mg intramuscularly

Dosage Forms

- Tablet 2 mg, 4 mg, 8 mg, 16 mg
- Injection 5 mg/mL (discontinued in USA)

How to Dose

- Oral: psychosis: 4–8 mg 3 times a day; 8–16 mg 2 times a day to 4 times a day in hospitalized patients; maximum 64 mg/day
- Oral: nausea/vomiting: 8–16 mg/day in divided doses; maximum 24 mg/day
- Intramuscular: psychosis: initial 5 mg; can repeat every 6 hours; maximum 15 mg/day (30 mg/day in hospitalized patients)



Dosing Tips

- Injection contains sulfites that may cause allergic reactions, particularly in patients with asthma
- Oral perphenazine is less potent than the injection, so patients should receive equal or higher dosage when switched from injection to tablet
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term

(one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly

- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Drug-induced parkinsonism, coma, hypotension, sedation, seizures, respiratory depression

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Slow down-titration of oral formulation (over 6–8 weeks), especially when simultaneously beginning a new antipsychotic

while switching (i.e., cross-titration)

- Rapid oral discontinuation may lead to rebound psychosis and worsening of symptoms
- If antiparkinson agents are being used, they should be continued for a few weeks after perphenazine is discontinued

Pharmacokinetics

- Half-life approximately 9.5 hours



Drug Interactions

- May decrease the effects of levodopa, dopamine agonists
- May increase the effects of antihypertensive drugs except for guanethidine, whose antihypertensive actions perphenazine may antagonize
- Additive effects may occur if used with CNS depressants
- Anticholinergic effects may occur if used with atropine or related compounds
- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome
- Epinephrine may lower blood pressure; diuretics and alcohol may increase risk of hypotension



Other Warnings/Precautions

- If signs of neuroleptic malignant syndrome develop, treatment should be immediately discontinued
- Use cautiously in patients with respiratory disorders
- Use cautiously in patients with alcohol withdrawal or convulsive disorders because of possible lowering of seizure threshold
- Do not use epinephrine in event of overdose as interaction with some pressor agents may lower blood pressure
- Avoid undue exposure to sunlight
- Avoid extreme heat exposure
- Use with caution in patients with respiratory disorders, glaucoma, or urinary retention
- Antiemetic effect of perphenazine may mask signs of other disorders or overdose; suppression of cough reflex may cause asphyxia
- Observe for signs of ocular toxicity (corneal and lenticular deposits)
- Use only with caution if at all in Parkinson's disease or Lewy body dementia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy

- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is in a comatose state or has CNS depression
- If there is the presence of blood dyscrasias, subcortical brain damage, bone marrow depression, or liver disease
- If there is a proven allergy to perphenazine
- If there is a known sensitivity to any phenothiazine

Special Populations

Renal Impairment

- Use with caution

Hepatic Impairment

- Use with caution; may not be recommended as long-term treatment because perphenazine may increase risk of further liver damage

Cardiac Impairment

- Cardiovascular toxicity can occur, especially orthostatic hypotension

Elderly

- Lower doses should be used and patient should be monitored closely

- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events

Children and Adolescents

- Not recommended for use in children under age 12
- Over age 12: if given intramuscularly, should receive lowest adult dose
- Generally consider second-line after atypical antipsychotics

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during

the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding

- Reports of drug-induced parkinsonism, jaundice, hyperreflexia, hyporeflexia in infants whose mothers took a phenothiazine during pregnancy
- Perphenazine should only be used during pregnancy if clearly needed
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Atypical antipsychotics may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy

Breast Feeding

- Unknown if perphenazine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

The Art Of Psychopharmacology

Potential Advantages

- Intramuscular formulation for emergency use

Potential Disadvantages

- Patients with tardive dyskinesia
- Children
- Elderly

Primary Target Symptoms

- Positive symptoms of psychosis
- Motor and autonomic hyperactivity
- Violent or aggressive behavior

Pearls

- Some studies suggest comparable effectiveness with atypical antipsychotics
- Perphenazine is a higher potency phenothiazine
- Less risk of sedation and orthostatic hypotension but greater risk of drug-induced parkinsonism than with low-potency phenothiazines
- Patients have very similar antipsychotic responses to any conventional antipsychotic, which is different from atypical antipsychotics where antipsychotic responses of individual patients can occasionally vary greatly from one atypical antipsychotic to another
- Patients with inadequate responses to atypical antipsychotics may benefit from a trial of augmentation with a conventional

antipsychotic such as perphenazine or from switching to a conventional antipsychotic such as perphenazine

- However, long-term polypharmacy with a combination of a conventional antipsychotic such as perphenazine with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy
- Availability of alternative treatments and risk of tardive dyskinesia make utilization of perphenazine for nausea and vomiting a short-term and second-line treatment option

Suggested Reading

Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of

adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

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Lieberman JA , Stroup TS , McEvoy JP , et al. Effectiveness of antipsychotic drugs in patients with chronic schizophrenia . *N Engl J Med* 2005 ;**353** (12):1209 –23.

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Quraishi S , David A . Depot perphenazine decanoate and enanthate for schizophrenia . *Cochrane Database Syst Rev* 2000 ;(2):CD001717 .

Phenelzine

Therapeutics

Brands

- Nardil
- Nardelzine
- *see index for additional brand names*

Generic?

Yes

Class

- Neuroscience-based Nomenclature: serotonin, norepinephrine, dopamine enzyme inhibitor (SN-EI)
- Monoamine oxidase inhibitor (MAOI)

Commonly Prescribed for

(bold for FDA approved)

- **Depressed patients characterized as “atypical,” “nonendogenous,” or “neurotic”**
- Treatment-resistant depression
- Treatment-resistant panic disorder
- Treatment-resistant social anxiety disorder

How the Drug Works

- Irreversibly blocks monoamine oxidase (MAO) from breaking down norepinephrine, serotonin, and dopamine
- This boosts noradrenergic, serotonergic, and dopaminergic neurotransmission

How Long Until It Works

- Onset of therapeutic actions usually not immediate, but often delayed 2–4 weeks following adequate dosing
- If it is not working within 6–8 weeks, it may require a dosage increase or it may not work at all
- May continue to work for many years to prevent relapse of symptoms

If It Works

- The goal of treatment is complete remission of current symptoms as well as prevention of future relapses
- Treatment most often reduces or even eliminates symptoms, but not a cure since symptoms can recur after medicine stopped
- Continue treatment until all symptoms are gone (remission)
- Once symptoms are gone, continue treating for 1 year for the first episode of depression
- For second and subsequent episodes of depression, treatment may need to be indefinite
- Use in anxiety disorders may also need to be indefinite

If It Doesn't Work

- Many patients have only a partial response where some symptoms are improved but others persist (especially insomnia, fatigue, and problems concentrating)
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Some patients who have an initial response may relapse even though they continue treatment, sometimes called “poop-out”
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider psychotherapy
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)
- Some patients may experience apparent lack of consistent efficacy due to activation of latent or underlying bipolar disorder, and require antidepressant discontinuation and a switch to a mood stabilizer



Best Augmenting Combos for Partial Response or Treatment Resistance

- * Augmentation of MAOIs has not been systematically studied, and this is something for the expert, to be done with caution and with careful monitoring
- * A stimulant such as d-amphetamine or methylphenidate (with caution; may activate bipolar disorder and suicidal ideation; may elevate blood pressure)

- Lithium
- Mood-stabilizing anticonvulsants
- Atypical antipsychotics (with special caution for those agents with monoamine reuptake blocking properties, such as lurasidone, ziprasidone, and zotepine)

Tests

- Patients should be monitored for changes in blood pressure; check before and 45–60 minutes after dosing until stable
- For patients receiving high doses or long-term treatment, consider periodically evaluating hepatic function
- * Since some MAOIs are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)
- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management
- * Monitor weight and BMI during treatment
- * While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or

dyslipidemia, or consider switching to a different antidepressant

Side Effects

How Drug Causes Side Effects

- Theoretically due to increases in monoamines in parts of the brain and body and at receptors other than those that cause therapeutic actions (e.g., unwanted actions of serotonin in sleep centers causing insomnia, unwanted actions of norepinephrine on vascular smooth muscle causing hypotension, etc.)
- Side effects are generally immediate, but immediate side effects often disappear in time

Notable Side Effects

- Dizziness, sedation, headache, sleep disturbances, fatigue, tremor
- Constipation, dry mouth, nausea, change in appetite, weight gain
- Sexual dysfunction (may be highest of any MAOI)
- Orthostatic hypotension (dose-related); syncope may develop at high doses
- Peripheral edema



Life-Threatening or Dangerous Side Effects

- Hypertensive crisis (especially when MAOIs are used with certain tyramine-containing foods or prohibited drugs)

- Induction of mania in patients with bipolar disorder
- Rare seizures
- Rare hepatotoxicity

Weight Gain



- Problematic; may be most of any MAOI

Sedation



- Many experience and/or can be significant in amount
- Can also cause activation

What to Do About Side Effects

- Wait
- Wait
- Wait
- Lower the dose
- Take at night if daytime sedation
- Switch after appropriate washout to an SSRI or newer antidepressant

Best Augmenting Agents for Side Effects

- Trazodone (with caution) for insomnia

- Benzodiazepines for insomnia

Dosing and Use

Usual Dosage Range

- 45–75 mg/day

Dosage Forms

- Tablet 15 mg

How to Dose

- Initial 45 mg/day in 3 divided doses; increase to 60–90 mg/day; after desired therapeutic effect is achieved lower dose as far as possible

Dosing Tips

- Once dosing is stabilized, some patients may tolerate once or twice daily dosing rather than 3 times a day dosing
- Orthostatic hypotension, especially at higher doses, may require splitting into 4 daily doses, or tincture of time (waiting)
- Patients receiving high doses may need to be evaluated periodically for effects on the liver
- Little evidence to support efficacy of phenelzine below doses of 45 mg/day

Overdose

- Death may occur; dizziness, ataxia, sedation, headache, insomnia, restlessness, anxiety, irritability, cardiovascular effects, confusion, respiratory depression, coma

Long-Term Use

- Consider periodic evaluation of hepatic function
- MAOIs may lose efficacy long-term

Habit Forming

- Dependence to MAOIs reported but rare

How to Stop

- Generally no need to taper, as the drug wears off slowly over 2–3 weeks

Pharmacokinetics

- Clinical duration of action may be up to 14 days due to irreversible enzyme inhibition



Drug Interactions

- Tramadol (serotonin reuptake inhibitor) may increase the risk of seizures in patients taking an MAOI
- Can cause a fatal “serotonin syndrome” when combined with drugs that block serotonin reuptake, so do not use with a serotonin reuptake

inhibitor or for 5 half-lives after stopping the serotonin reuptake inhibitor (see [Table 1](#) after Pearls)

- Hypertensive crisis with headache, intracranial bleeding, and death may result from combining MAOIs with sympathomimetic drugs (e.g., amphetamines, methylphenidate, cocaine, dopamine, epinephrine, norepinephrine)
- Do not combine with another MAOI, alcohol, or guanethidine
- Adverse drug reactions can result from combining MAOI with tricyclic/tetracyclic antidepressants and related compounds, including carbamazepine, cyclobenzaprine, and mirtazapine, and should be avoided except by experts
- MAOIs in combination with spinal anesthesia may cause combined hypotensive effects
- Combination of MAOIs and CNS depressants may enhance sedation and hypotension

Table 1. Drugs contraindicated due to risk of serotonin syndrome/toxicity

Do Not Use:			
Antidepressants	Drugs of Abuse	Opioids	Other
SSRIs	MDMA (ecstasy)	Meperidine	Non-subcutaneous sumatriptan
SNRIs	Cocaine	Tramadol	Chlorpheniramine
Clomipramine	Methamphetamine	Methadone	Brompheniramine

Do Not Use:

Antidepressants	Drugs of Abuse	Opioids	Other
Imipramine	High-dose or injected amphetamine		Dextromethorphan
St. John's wort			Lumateperone Ziprasidone

 Other Warnings/Precautions

- Use requires strict adherence to low-tyramine diet (see [Table 2](#) after Pearls)
- Patient and prescriber must be vigilant to potential interactions with any drug, including antihypertensives and over-the-counter cough/cold preparations
- Over-the-counter medications to avoid include cough and cold preparations, including those containing dextromethorphan (serotonin reuptake inhibitor), nasal decongestants (tablets, drops, or spray), hay-fever medications, sinus medications, asthma inhalant medications, anti-appetite medications, weight reducing preparations, “pep” pills (see [Table 3](#) after Pearls)
- Hypoglycemia may occur in diabetic patients receiving insulin or oral antidiabetic agents (hydrazine effect)
- Binds and inactivates vitamin B6 (hydrazine effect); reversible with high-dose B6 administration

- Use cautiously in patients receiving reserpine, anesthetics, disulfiram, metrizamide, anticholinergic agents
- Phenelzine is not recommended for use in patients who cannot be monitored closely

Table 2. Dietary guidelines for patients taking MAOIs*

Foods to avoid**	Foods allowed
Dried, aged, smoked, fermented, spoiled, or improperly stored meat, poultry, and fish	Fresh or processed meat, poultry, and fish; properly stored pickled or smoked fish
Broad bean pods	All other vegetables
Aged cheeses	Processed cheese slices, cottage cheese, ricotta cheese, yogurt, cream cheese
Tap and unpasteurized beer	Canned or bottled beer and alcohol
Marmite	Brewer's and baker's yeast
Banana peel	Bananas, avocados, raspberries
Sauerkraut, kimchee	Peanuts
Soy products/tofu	
Tyramine-containing nutritional supplement	

* Consult up-to-date listing of tyramine content, as most foods now have low tyramine amounts

** Not necessary for 6 mg transdermal or low-dose oral selegiline

Table 3. Drugs that boost norepinephrine: should only be used with caution with MAOIs

Use With Caution:			
Decongestants	Stimulants	Antidepressants with norepinephrine reuptake inhibition	Other
Phenylephrine	Amphetamines	Most tricyclics	Phentermine
Pseudoephedrine	Methylphenidate Cocaine Methamphetamine Modafinil Armodafinil	NRIs NDRI	Local anesthetics containing vasoconstrictors Tapentadol

Do Not Use

- If patient is taking meperidine (pethidine) (serotonin reuptake inhibitor)
- If patient is taking a sympathomimetic agent or taking guanethidine
- If patient is taking another MAOI
- If patient is taking any agent that can inhibit serotonin reuptake (e.g., SSRIs, tramadol, milnacipran, duloxetine, venlafaxine, clomipramine, imipramine, etc.)